SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Total Marks: 50

Code of Conduct

I K. Sasidhar Reddy (192525140) (Name / Reg No)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K. Sasidhar Reddy.

Assessment Tasks

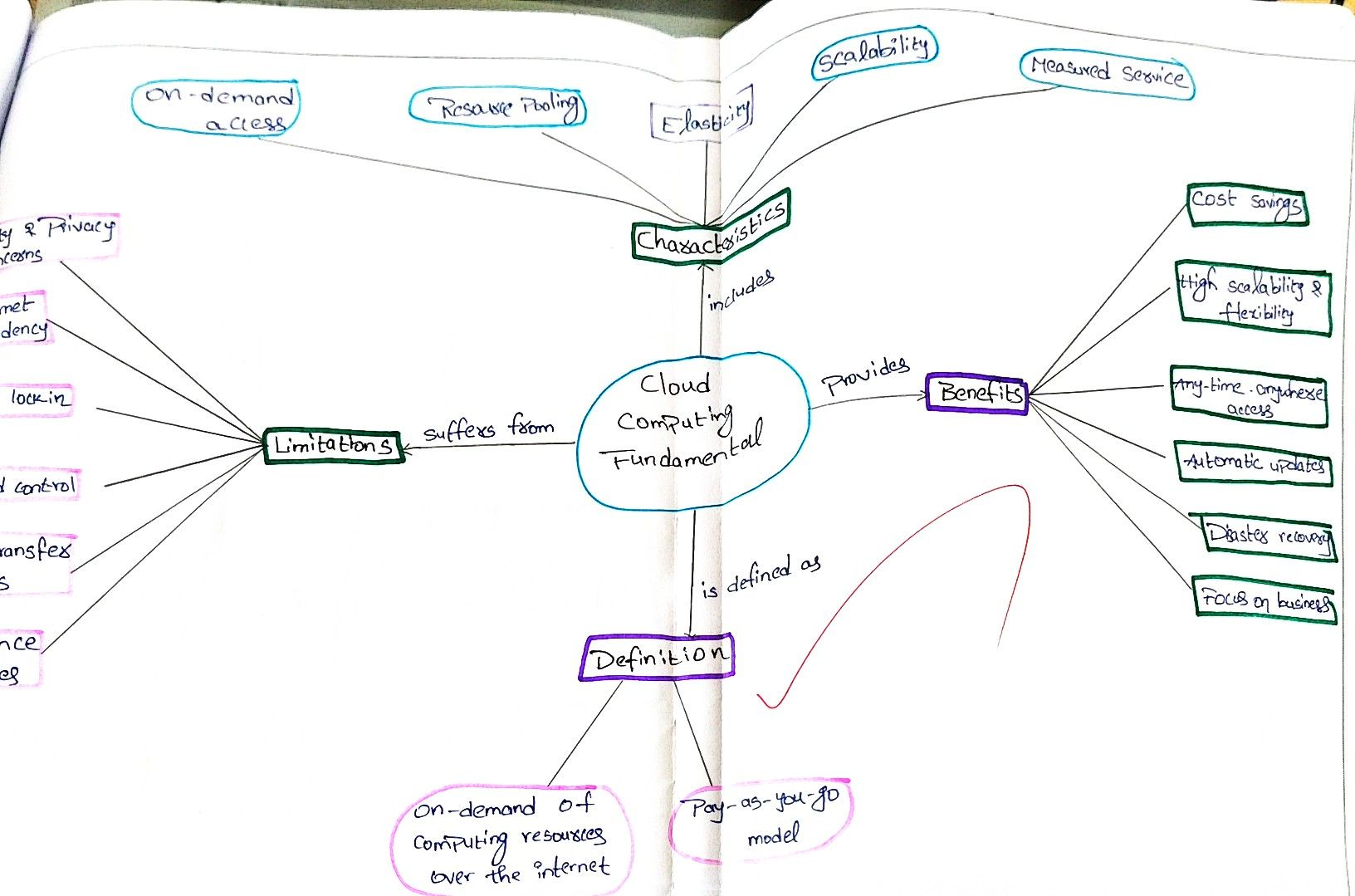
Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

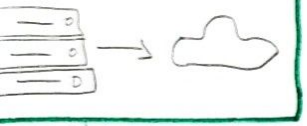
Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility


Cloud Computing Fundamental
Characteristics
includes
On-demand access
Resource Pooling
Elasticity
Scalability
Measured Service
Limitations
suffers from
y & Privacy
rcerns
met
dency
lock-in
l control
ransfer
s
nce
es
Provides
Benefits
Cost savings
High scalability & flexibility
Any-time-anywhere access
Automatic updates
Disaster recovery
Focus on business
is defined as
Definition
On-demand of computing resources over the internet
Pay-as-you-go model

Evolution of cloud computing

Hardware Evolution



- Mainframes (1st Gen)
- Minicomputers (2nd Gen)
- Personal Computers (3rd Gen)
- Virtualized / Modern servers 4th Gen

Internet software Evolution



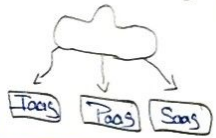
- Static web (1990s)
- Dynamic web (2000s)
- web 2.0 (Social Media, Collaboration)
- web 3.0 (Cloud, AI, IOT)

Distributed System

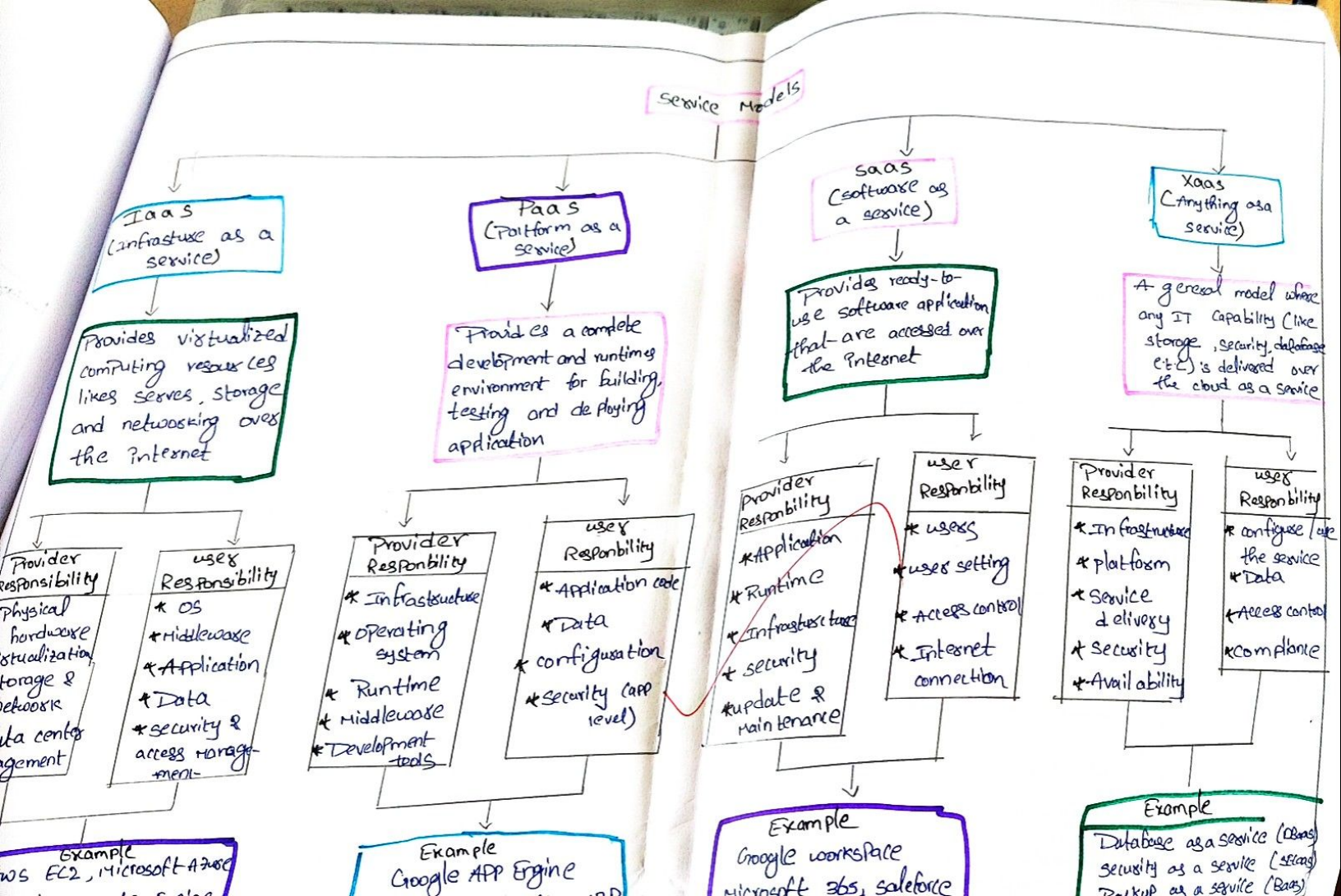


- Multiple computers working together
- Data sharing
- Fault tolerance
- High availability

Service Delivery



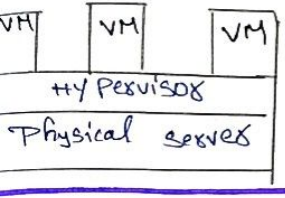
- Infrastructure as a service
- Platform as a service
- Software as a service
- Xaas (other services)


Service Models
Iaas (infrastuse as a service)
Provides virtualized computing resources likes serves, storage and networking over the Internet
Provider Responsibility
Physical hardware
virtualization
storage &
networks
data center
agement
user Responsibility
* OS
* Middleware
* Application
* Data
* security & access manage-ment-
Example
ws EC2, Microsoft Azure
Paas (Platform as a service)
Provides a complete development and runtime environment for building, testing and deploying application
Provider Responsibility
* Infrastructure
* operating system
* Runtime
* Middleware
* Development tools
user Responsibility
* Application code
* Data
* configuration
* Security (app level)
Example
Google APP Engine
saas (software as a service)
Provides ready-to-use software application that are accessed over the Internet
Provider Responsibility
* Application
* Runtime
* Infrastructure
* security
* update & maintenance
user Responsibility
* users
* user setting
* Access control
* Internet connection
Example
Google workspace
Microsoft 365, saleforce
Xaas (Anything as a service)
A general model where any IT capability (like storage, security, database etc) is delivered over the cloud as a service
Provider Responsibility
* Infrastructure
* platform
* Service delivery
* security
* Availability
user Responsibility
* configure / use the service
* Data
* Access control
* Compliance
Example
Database as a service (DBaas)
security as a service (SECaas)
Backup as a service (Baas)

Virtualization and web services as Enablers of cloud

virtualization

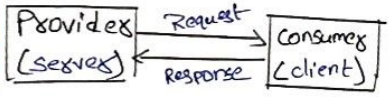
creates virtual machines (VMs) on a physical server.
managed by a hypervisor
Provides abstraction of hardware resources



Cloud Deployment

web services

* Enables communication between application
* uses standard Protocols (SOAP / REST)
* Supports Providers and Consumer Interaction



12/10/26

Cloud Deployment

* Easy Integration
* Interoperability

scalability

* Add/remove resources
* Handle varying demand
* Better performance

Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented

Total Marks: